

SpaceX Falcon 9 First-Stage Landing Prediction

IBM Data Science Capstone Project

Final presentation | Data collection, analysis, interactive visualization, and machine learning

Dataset period: 2010-2020

The analysis identifies where and when Falcon 9 landings succeed

A reusable first stage lowers launch cost, so predicting landing success supports competitive bid estimates.

90

Falcon 9 launches

87.8%

successful landings in
prepared labels

83.3%

test accuracy across four
models

41.7%

share of all successful
launches at KSC LC-39A

Key implication

Operational history and launch context provide useful predictive signal, while launch site, orbit, payload, and booster configuration explain meaningful differences in outcomes.

The project combines seven analytical stages



Evidence was retained in executed notebooks, dashboard screenshots, and an interactive Folium HTML map.

Two collection paths build the launch record

SpaceX REST API

- Past-launch JSON normalized into a DataFrame
- Rocket, payload, core, and launchpad identifiers enriched
- Final Falcon 9 table contains 90 launches and 17 fields

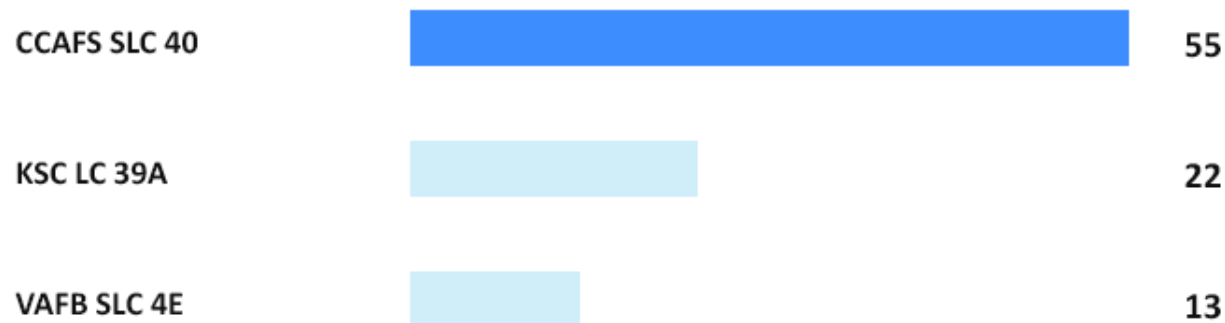
Wikipedia web scraping

- BeautifulSoup parses historical launch tables
- 121 launch rows extracted from the June 2021 snapshot
- Fields include booster, payload, orbit, customer, outcome, and landing

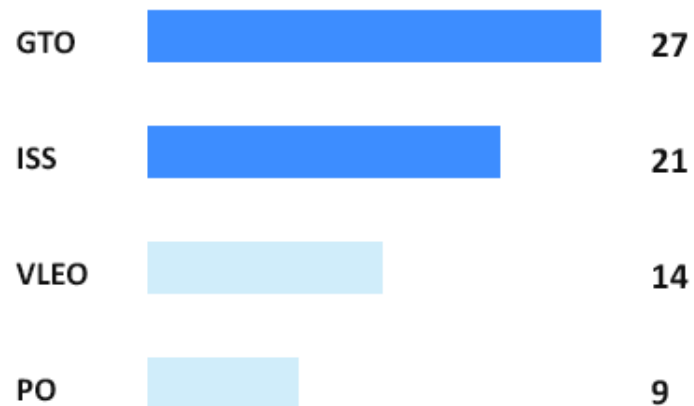
Both paths provide complementary operational and mission context.

Launch activity concentrates at three sites and a few orbit types

Launches by site

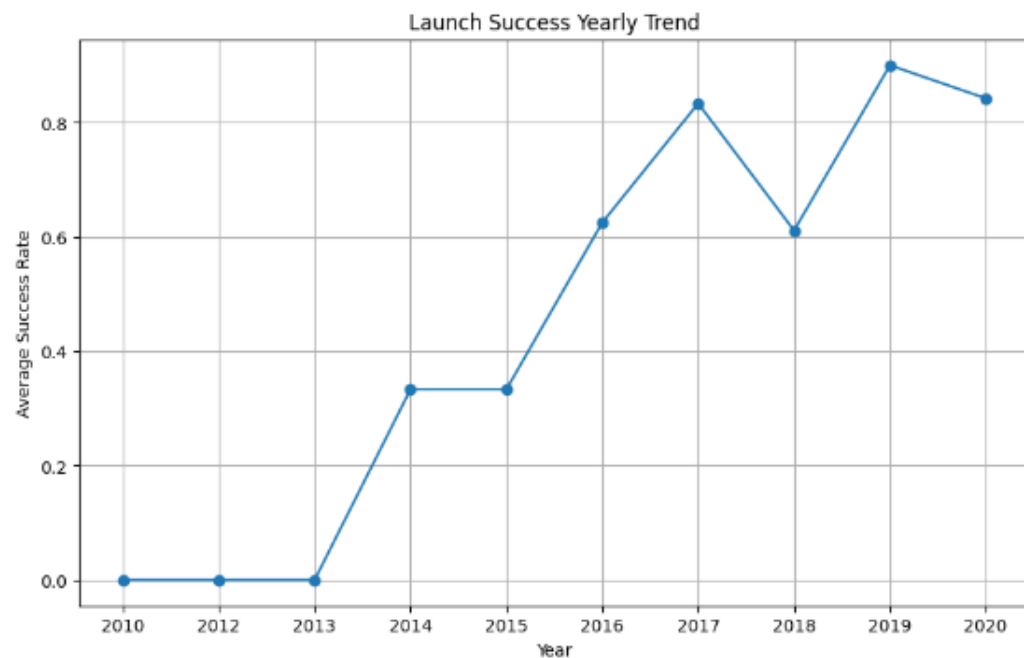


Most common orbits

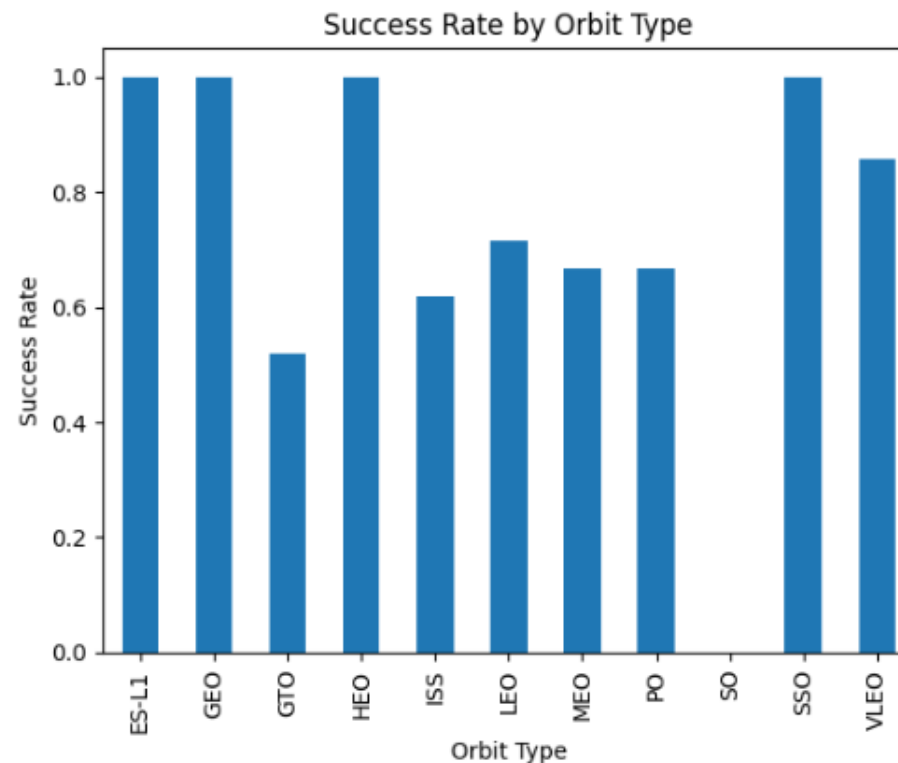


Landing outcomes were converted to a binary Class label; the resulting success share is 87.8%.

Success improves over time and varies by orbit



The yearly trend rises as Falcon 9 operations mature.



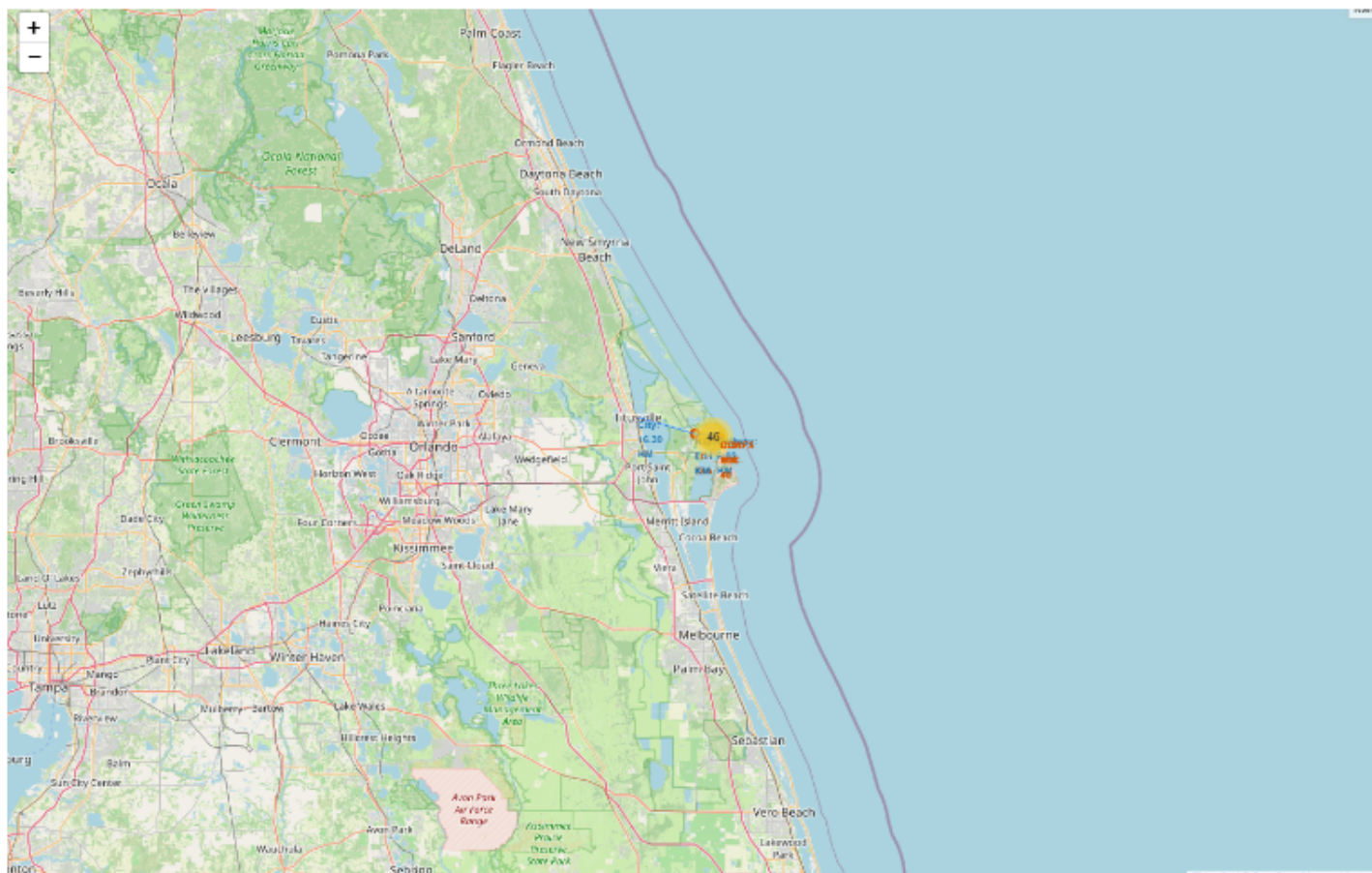
Orbit choice separates high- and low-success mission profiles.

SQL confirms operational milestones and payload patterns

NASA CRS payload	45,596 kg
Avg. F9 v1.1 payload	2,928.4 kg
First ground-pad success	22 Dec 2015
Successful missions	100*
Drone-ship successes, 2010-2017	5

*SQL source records include "Success" and "Success " as separate text values.

Launch sites cluster on coastlines with transport access

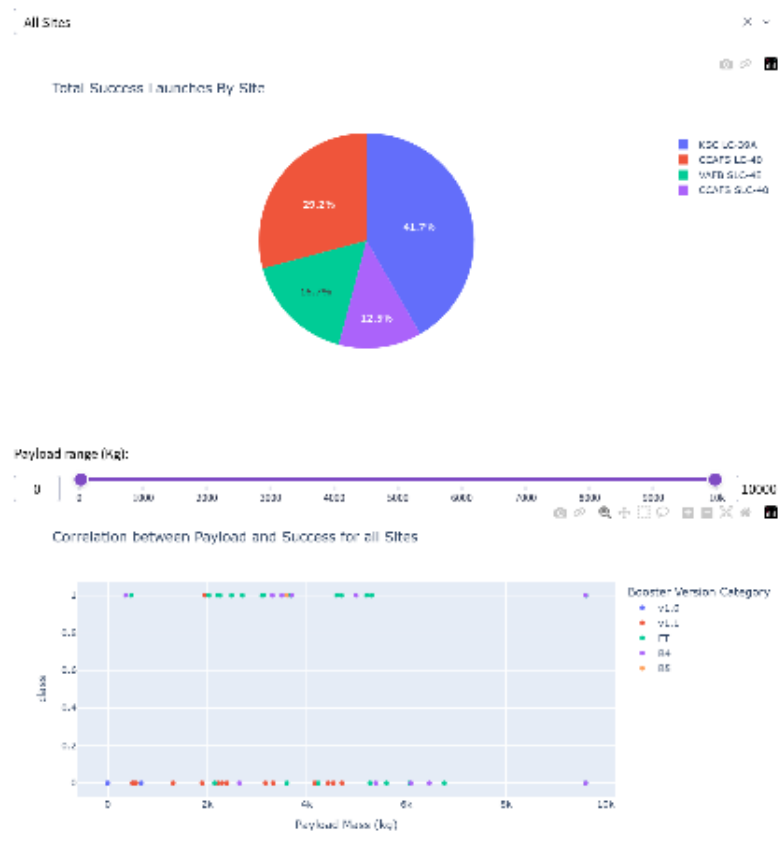


Geographic findings

- Sites are near coastlines, limiting risk over populated areas.
- Florida sites cluster around Cape Canaveral and Kennedy Space Center.
- The interactive map marks launch outcomes and distances to coast, city, railway, and highway.

KSC LC-39A contributes the largest share of successful launches

SpaceX Launch Records Dashboard

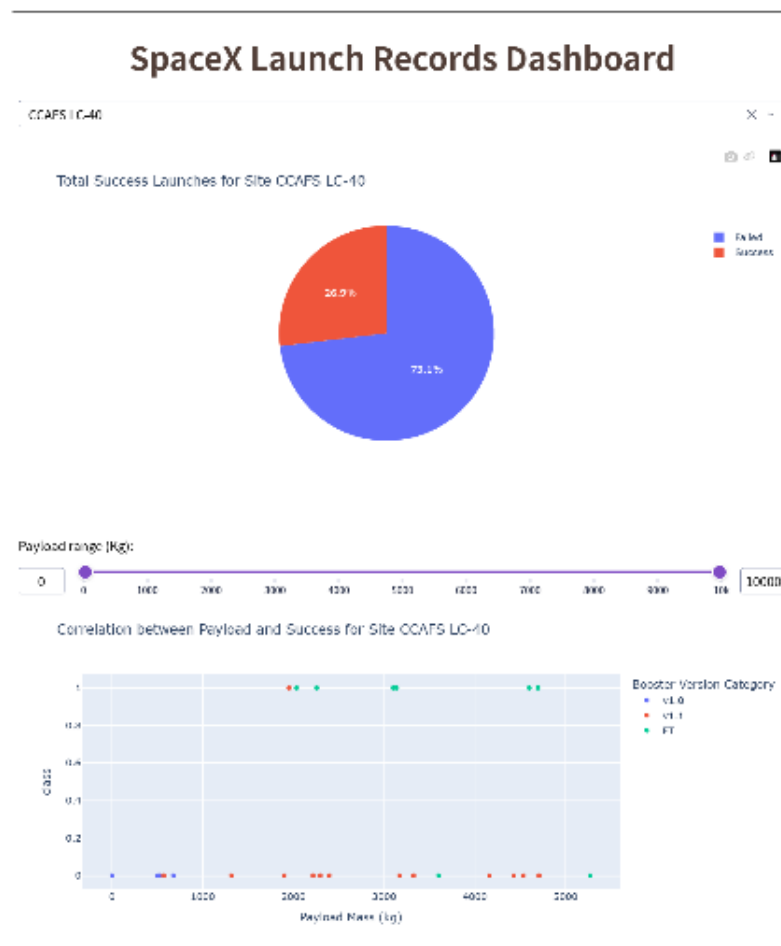


41.7%

of all successful launches
occurred at KSC LC-39A

The payload scatter also shows
outcome differences across
booster versions.

CCAFS LC-40 records a 26.9% success rate



26.9%

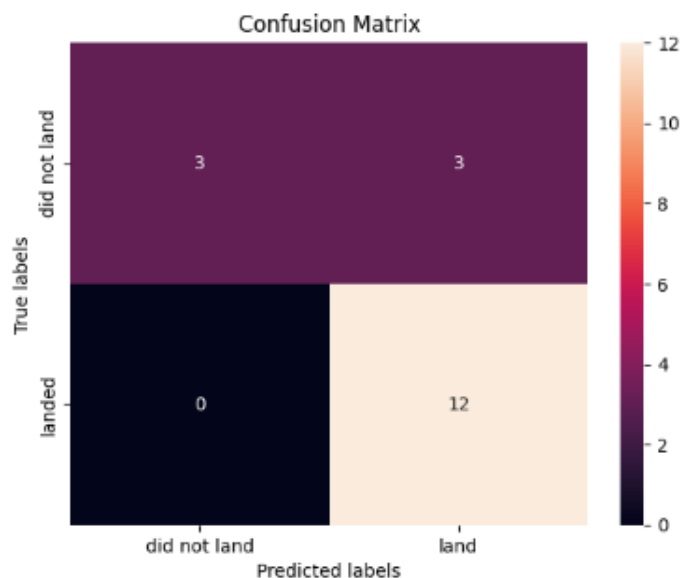
successful launches

73.1% failed

Filtering by site changes both the outcome pie and payload scatter, demonstrating the dashboard's interactivity.

All four classifiers reach 83.3% test accuracy

Model	CV accuracy	Test accuracy
Logistic regression	84.6%	83.3%
Support vector machine	84.8%	83.3%
Decision tree	92.9%	83.3%
K-nearest neighbors	84.8%	83.3%



Decision tree leads cross-validation, but identical test scores indicate that the 18-case test set is too small to establish a clear winner.

Landing success reflects operational maturity and mission context

- | | | |
|----|--|--|
| 01 | Performance improves over time | Later flights show stronger landing outcomes as operations mature. |
| 02 | Site and orbit matter | KSC leads successful launches; outcome rates differ across sites and orbit profiles. |
| 03 | Reusable hardware features carry signal | Grid fins, legs, reuse history, block, orbit, and payload contribute to prediction. |
| 04 | Models are useful but need broader validation | 83.3% test accuracy is promising, though the test set contains only 18 launches. |

Submission package maps directly to the capstone rubric

Included

- API collection notebook
- Web scraping notebook
- Data wrangling notebook
- EDA visualization notebook
- SQL notebook
- Folium notebook + HTML map
- Dash application + screenshots
- Machine learning notebook

Public GitHub repository

All completed notebooks, dashboard code, screenshots, map, presentation, and report are available in the repository.

<https://github.com/TristinWen/spacex-falcon9-landing-prediction>